

Attendance Recording System Using Deep Face Detection and Recognition Models

1st Enais Adnan Moses Deli
Department of Computer Science, College of
Education, University of Kufa
Najaf, Iraq
enaisa.abughoneim@student.uokufa.edu.iq

2nd Hasan Thabit Rashid Kurmasha
Department of Computer Science, College of Education for
Women, University of Kufa
Najaf, Iraq
hasant.kurmasha@uokufa.edu.iq

Abstract—In this paper, deep neural network called “Yunet” is used for face detection and the sigmoid-constrained hypersphere loss called “Sface” is applied for face recognition to track student attendance in real educational institutions. Both Yunet and Sface common in using five landmarks for each face which is a new and effective way to improve the accuracy, efficiency over attendance management systems through dynamic and different sizes of deep facial detection and recognition with high accuracy and faster image processing. Three datasets are used in; the first one is designated and generated on educational institutions classes for academy students and the second one for secondary stage school were the recorded videos are collected in real standard and tight criteria on lightning, distances, faces directions, and the angles of leans while the third one is a generated video for a popular actors’ images dataset. The designated system is developed and programed in Python and OpenCV in three phase’s starting from face detection followed by face recognition and ending with attendance report collection in excel file with its metadata. However, the results show that both Yunet and Sface models can avoid all difficulties and gotten an excellent accuracy each alone 100% and 98.14% respectively while the overall attendance recording system has average accuracy 100% in recording the student status whether it was present or absence by dealing with tough conditions. Finally, such this automated system will definitely eliminate the traditional ways drawbacks for record attendance manually.

Keywords—Attendance recording systems, Face detection and recognition, DNN, CNN, Yunet model, Sface model.

I. INTRODUCTION

In educational institutions like schools and colleges, an attendance system is used to automatically record and track student attendance, by eliminating the need prone manual signing or roll calling. The system seeks to increase efficiency and accuracy in the recording of attendance [1],[2]. Manual attendance recording takes a lot of time and effort from teachers. It is also prone to error when recording names, and it may be used to record someone else's attendance. Thus, the demand need for an automatic attendance recording system that is effective, dependable, and user-friendly to track attendance in schools [3],[4],[5],[6],[7],[8]. This technology analyzes face features like the mouth, nose, and eyes and turns them into digital data that may be compared to images kept in a database to enable identification [9],[10]. Facial recognition is an excellent approach to identify persons without the need for manual contact, which helps reduce manipulation in the attendance system [11]. Yunet model was utilized in recent works for deep facial detection and however make high accuracy and quick image processing over attendance

tracking [12] while Sface model is used to build more resilient face recognition models [13]. These models was developed using OpenCV and Python which offers a comprehensive library for image processing techniques [14]. The other sections of this paper are as following; explain the most significant studies and techniques pertaining to attendance recording, and talk about the advantages and disadvantages of these techniques in the related works section. The third section shows the general flowchart of the proposed system approach and all details about applying it. Forth section presents the instruments and methods employed and the experimental strategy for design and data collection as well as present the results and findings appropriately with explanation using tables and graphs. The conclusion explains the work contribution and provides a future work of the key findings to the study. The work contribution can be summarized as following; first, designing attendance recording system based on efficient DNN models called Yunet, Sface. Second, designing and use two datasets from two educational institutions in real environment. Finally, excellent results are gotten and evaluated subjectively and objectively.

II. RELATED WORKS

Researchers The most popular techniques used for attendance recording systems with the advantages and disadvantages can be described as following. *FaceNet* is a face recognition model based on deep learning techniques (DNN). It converts a face image into a digital vector that can be used for easy or identification by comparing the distances between the vectors. *FaceNet* has proven its performance on popular databases such as Labelled Faces in the Wild (LFW) and achieved an accuracy of up to 99.63%. To get the best results, a lot of data and training are needed. However, it can be enhanced by better data management and less reliance on computational resources [15]. *Region-based Convolutional Neural Network (R-CNN)* is a sophisticated CNN-based face detection method. Applied for many huge datasets like WIDER FACE, FDDB, and IJB-A to provide fast and accurate results and it has demonstrated excellent performance on numerous face identification benchmarks. Nevertheless, it still requires more computer resources than simpler models. To enhance performance under various circumstances, a substantial and varied amount of training data is needed [16], [17]. *Retina Face* is a new face detection and recognition technique that uses a multi-level approach to accurately detect faces and identify landmarks in images as in Fig. 1. The technique's ability to find faces, predict 2D facial landmarks, and rebuild 3D faces is typified by its use of single-shot images. In order to extend various

face ranges, the system incorporates features from hierarchical networks and is built on CNN technology. [18], [19]. *You Only Look Once (YOLO)* is a deep Learning model used to tackle challenging tasks like rapidly and effectively identifying faces and objects in photos. Datasets like WIDER FACE and Celeb-Faces have been used to test the model. Even in huge images, YOLO has demonstrated strong performance, recognizing faces in 0.027 seconds. Due to its inability to adjust to fine features, it may have trouble detecting little or distant faces, but it is resilient in complicated surroundings and consistently performs well even in trying circumstances. Significant resources are needed to provide optimal performance [20],[21],[22].

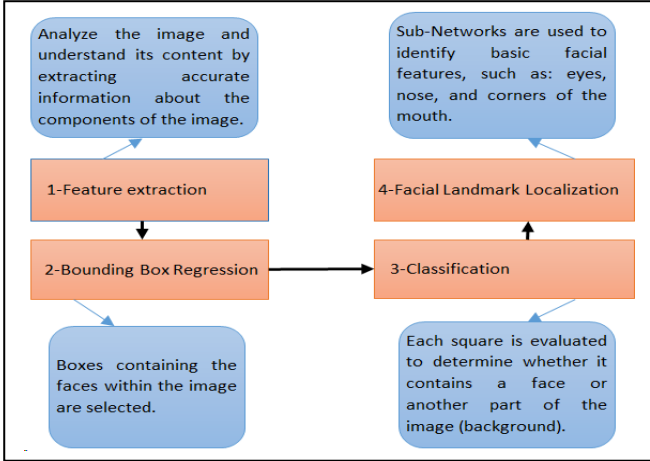


Fig. 1. The general steps of Retina Face model.

The new Yunet model is a lightweight and effective face detection model. Its main goal is to strike a superb balance between accuracy, speed, and efficiency. It performs similarly to larger models on datasets like WIDER FACE and uses less memory and power, making it appropriate for real-time applications. It is recently used to solve the problem of heavy and computationally expensive models that cannot be easily deployed on devices with limited computing capabilities. It is intended to keep a high degree of accuracy while lowering latency and parameter counts. There are three primary components to the model [12] as in Fig. 2.

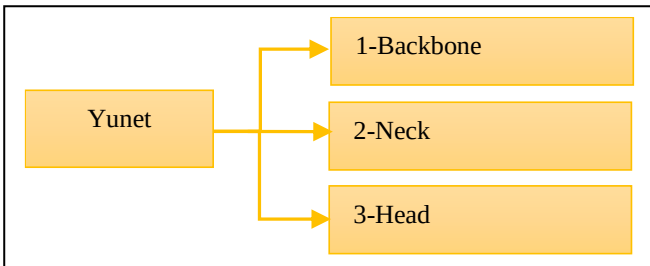


Fig. 2. The general flowchart of the Yunet model comonents.

As to Fig. 2 above, the first one is backbone which is used to extract features from the input image for analysis in the detection phases. It focuses on small face detection, which is more difficult than large face detection. The second one is neck which is a crucial part and incorporates characteristics from several levels into the backbone in order

to increase the model's ability to detect faces of various sizes and situations (such as small and large faces). The final is the head which detect faces using the multi-level features that Neck has built in. Head concentrates on delivering precise detection outcomes. The results of applying the Yunet model as compared to other ones are as to the following Table 1. Yunet has the fewest parameters 75,856 and is ten times smaller than the majority of other models in size. When it comes to inference time, Yunet outperforms other models by many times.

TABLE I. YUNET COMPARISON WITH OTHER POPULAR TECHNIQUES [12].

Input size	Methods	Parameters (ratio)	FLOP (M)	APhead	Latency (ms)
320*320	RetinaFace	27293600 (359.81x)	11070	0.341	49.1
	SCRFD-10g	4229905 (55.76x)	3359	0.504	17.3
	Yunet	75856 (1.00x)	149	0.395	2.2
640*640	RetinaFace	-	44260	0.659	232.7
	SCRFD-10g	-	13435	0.814	95.0
	Yunet	-	595	0.691	11.3
Origin size	RetinaFace	-	-	0.847	463.7
	SCRFD-10g	-	-	0.885	137.8
	Yunet	-	-	0.811	16.3

The new Sface model is a loss function used to training databases with low-quality photos and data noise (like misaligned or low-resolution images), which is used to build more resilient face recognition models. it employs sigmoid curves to rebalance gradients and optimizes intra-class and inter-class targets on a spherical manifold In order to enhance training on clean images and lessen the detrimental impacts of noise. Databases like MS-Celeb-1M, VGGFace2, and CASIA-Web-Face were used for training. When testing recognition using one million images, Sface achieved similar or better performance than other advanced methods, as shown in Table 2. However, more details can achieved in [13].

TABLE II. SFACE VERIFICATION PERFORMANCE WITH OTHER POPULAR TECHNIQUES BASED ON ON LFW AND YTF DATABASES [13].

Methods	Total Images in Million	LFW	YTF
FaceNet	200 M	99.63	95.10
ArcFace	5.8 M	99.83	98.02
Sface	5.8 M	99.82	98.06

III. THE PROPOSED ATTENDANCE SYSTEM

Several steps will be taken into account to accomplish the goal of creating an efficient facial attendance recording system using Yunet and Sface models as mentioned previously in related work section. Setting and designing two datasets in educational environment for collagists and school students as well as using the actor's dataset will mention that in section 4. The general flowchart of the proposed attendance recording system as in Fig. 3.

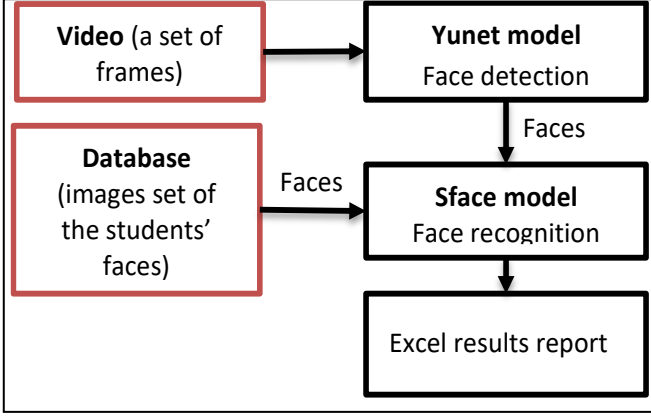


Fig. 3. The general flowchart of the proposal attendance recording system using Yunet and Sface models.

A. The Face Detection Stage

This stage is done using Yunet model after capturing the class videos, the faces in are discovered. the layers in Yunet analyze the image to look for distinctive patterns of faces and then output bounding boxes of the detected faces in the image as in following Fig. 4 and Fig. 5 respectively.

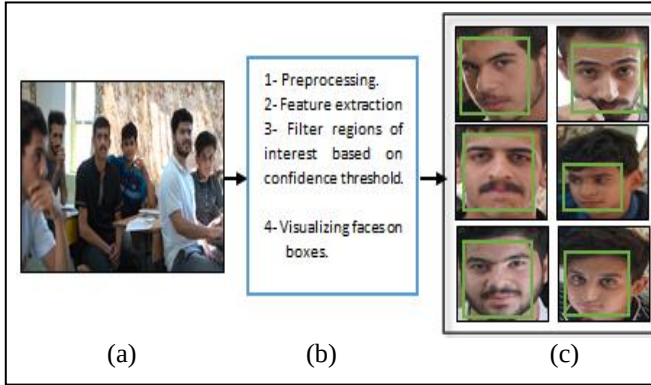


Fig. 4. The general steps of Yunet model. (a)current frame, (b)Yunet steps, (c) The faces detection by boxes.



Fig. 5. The faces boxes have basic five landmarks features from frame1, video3, School dataset.

One advantage is that model does not need training as it is well trained and effective based on its “.onnx feature file” of face detection and hence; high accuracy in detecting faces even in environments with poor conditions such lighting or the face is not ideal, indirect angles, etc.

B. The Face Recognition Stage

There are some things that affect the accuracy of recognition such as changes in lighting and angles, and the images may not be good or some student who may look very similar to others or his face is not clear or the image has noise which makes the recognition process difficult. The Sface model avoids that and hence; the students' faces are recognized in an excellent way even there are unsuitable conditions. The recognition process begins, which is matched the faces detected in the video' frame with the faces in the dataset. Here, a certain dynamic thresholds is used and based on this thresholds, the system considers that the face in the video matches one of the faces in the database as Fig. 6 and Fig. 7 respectively.

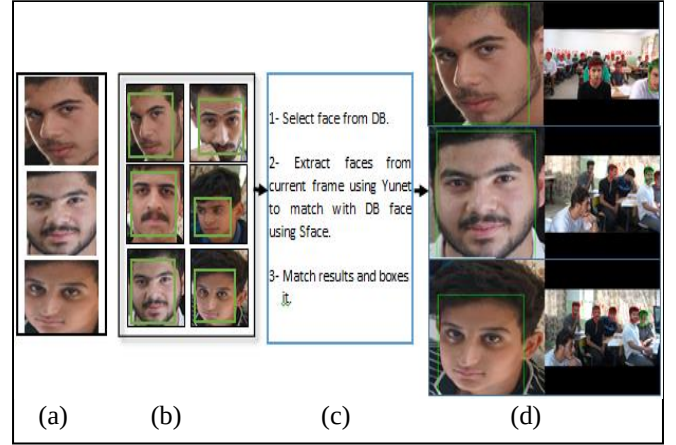


Fig. 6. The general steps of Sface model. (a)saved DB, (b)detected faces in current frame by Yunet model, (c) The Sface steps,(d) the face from DB and its location in the frame.

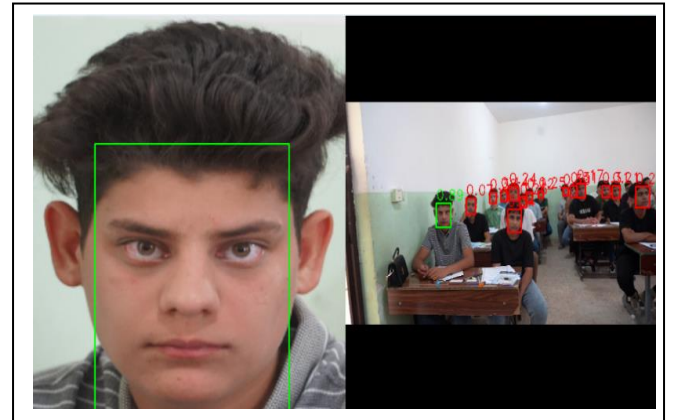


Fig. 7. The student face recognition using Sface model and dynamic thresholds from frame1, video3, School dataset.

C. Attendance Recording Stage

After the face recognition stage, attendance is recorded for faces that are recognized as present by putting the 1 value in report excel file, otherwise 0 for absence that are not recognized faces according to the applied thresholds in the

matching process. The final result is saved in an Excel sheet as a daily attendance sheet as Fig. 8.

number	name	result
1	abbas zaid ali	1
2	abdaz nafie jameel	1
3	abdullah ali dakhil	1
4	abdullah aziz jabbar	0
5	ahmed salem dahad	0
6	ali al-murtada hamad	1

Fig. 8. The report results of attendance recording system.

IV. RESULTS AND DISCUSSIONS

The results of applying the Yunet and Sface models and the proposed attendance recording system for the created datasets will be show in following sub-sections.

A. New Dataset Creation

As in real world educational environment, three cameras used (a ready camera, a computer camera, and a mobile camera) to recorded videos for the 36 volunteers' students for one minute. Two sessions in different times were taken in mathematics department of the faculty of education in Kufa University, after each session their seating arrangements were then altered, and a video was taken of them in order to alter the students' looks, positions, and faces directions and saving that in "**University**" folder, the re-arrangements of students' positions shown in Fig. 9. Additionally, two sessions in the same scenario were taken of the secondary school for 40 students and stored as a dataset in the "**School**" folder as shown in Fig. 10. Also, a valuable dataset on internet is used for selected 31 actors' dataset and 5 images for each actor were saved in a folder called **Actors** [23]. There are 100 faces in this folder as in Fig. 11. The remaining images containing the entire body of each actor were compiled as video to recognition stage.

P.1	P.5	P.9	P.13	P.11	P.15	P.3	P.7
P.2	P.6	P.10	P.14	P.12	P.16	P.4	P.8
P.3	P.7	P.11	P.15	P.9	P.13	P.1	P.5
P.4	P.8	P.12	P.16	P.10	P.14	P.2	P.6

(a)

(b)

Fig. 9. The students' positions in classroom. (a) First session, (b) second session (re-arrange the positions).

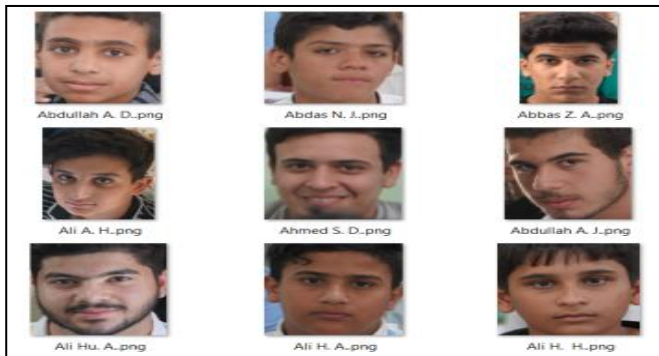


Fig. 10. The students faces of the images in the used School dataset.



Fig. 11. The actors' faces as sampled images of the used Actors dataset .

B. The Proposed System Applying

In order to apply the proposed system, a programed interface was designated and used to execute the attendance recording system as in the Fig. 12. The user need to select the video directory from the computer and then, apply the attendance recording button, the result will be directly reported to the excel file sheet. The results of using University dataset, 36 faces were recorded as present for all students in all videos, which mean all faces were detected and recognized entirely while in School dataset; 34 faces were recorded as present for all students in all videos which mean all faces were detected and recognized entirely as well as in Actor' dataset, 31 faces were detected and recognized entirely.

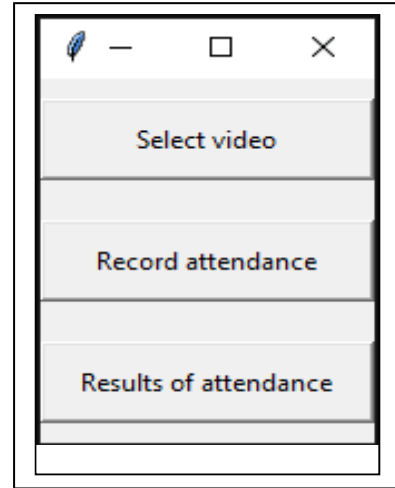


Fig. 12. The interface for execution of the proposed attendance recording system.

The following Table 3, Table 4, and Fig. 13 show the results of detection stage and recognition stage using the popular evaluation metrics recall, precision, F-score which counted based on true positive (TP) which is the number of faces correctly detected and/or recognized as the same face as in the database of images, true negative (TN) which is the number of images that are not faces and are correctly predicted as 'not a face' by the applied algorithm, false positive (FP) refer to the number of images that are not faces, but are classified as faces, and false negative (FN) refer to number of images that are faces, but not detected as faces.

TABLE III. THE AVERAGE ACCURACY OF FACE DETECTION STAGE USING YUNET MODEL IS 100%. SEE THAT THE MODEL OUTPERFORMS ALL CHALLENGES IN RECORDING VIDEOS CIRCUMSTANCE SUCH AS DISTANCES, VIBRATION, OCCLUSIONS, SHOOTING, ETC.

Dataset	student	Videos	Detected faces	TPR	FNR	Precision	Recall	Accuracy	F1-score
University	36	V1	36	36	0	1	100	100	100
		V2	36	36	0	1	100	100	100
		V3	36	36	0	1	100	100	100
School	34	V1	34	34	0	1	100	100	100
		V2	34	34	0	1	100	100	100
		V3	34	34	0	1	100	100	100
Actors	31	V1	31	31	0	1	100	100	100

TABLE IV. THE AVERAGE ACCURACY OF FACE RECOGNITION STAGE USING SFACE MODEL IS 98.14%.

Dataset	students	Videos	Recognize faces	TPR	FNR	Precision	Recall	Accuracy	F1-score
University	36	V1	35	35	1	1	0.97	0.97	0.98
		V2	33	33	3	1	0.91	0.91	0.95
		V3	36	36	0	1	100	100	100
School	34	V1	32	32	2	1	0.94	0.94	0.96
		V2	33	33	1	1	0.97	0.97	0.98
		V3	34	34	0	1	100	100	100
Actors	31	V1	31	31	0	1	100	100	100



Fig. 13. The Face detection using YUNET model in red boxes and the recognition faces using Sface model in green boxes for five students faces from the database in the first row. Frame5, Video3, School dataset.

As to above Table 3, all faces in the videos of university and school dataset, and the actors datasets were detected

under all conditions and the accuracy is excellently 100%. These results improve the power of Yunet model. In Table 4, because of the absence students as in FNR column, the results were scored 98.14 %. The results of attendance recording system in general, there are the present case and absence case that will be used to measure the accuracy of system as in Table 5 which achieved average accuracy 100%.

TABLE V. THE AVERAGE ACCURACY RESULTS OF THE ATTENDANCE RECORDING SYSTEM IS 100%.

Dataset	Total Students/Actors	Video	Present	Absent	Accuracy
University	36	V1	35	1	1.00
		V2	33	3	1.00
		V3	36	0	1.00
School	34	V1	32	2	1.00
		V2	33	1	1.00
		V3	34	0	1.00
Actor	31	V1	31	0	1.00

The result of the proposed attendance recording system applying for the created two university and school datasets plus the selected actors dataset can be represented in column chart as following Fig. 14.

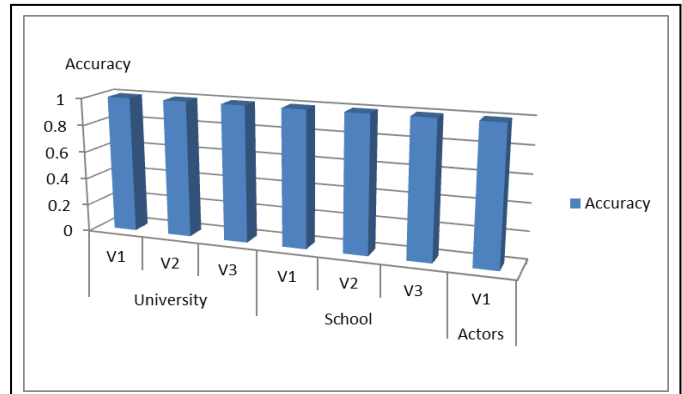


Fig. 14. The average Accuracy results 100% of the proposed attendance recording system based on the present and/or absence individuals for all videos of the three used datasets.

The experiments of recording videos has many challenges that affect the recognition process such as distances, occlusions, capturing false in the sessions, etc. therefore, to apply such attendance recording systems there is a need to detect perfect positions for placing the cameras and avoiding the vibrations when using hands for mobiles. Finally, the details for all applied datasets of each video in respect to durations, lengths, resolutions, sizes, and the execution times are shown in the following Table 6.

TABLE VI. THE VIDEOS LENGTH, RESOLUTIONS, SIZES, AND THE EXECUTION TIMES FOR ALL DATASETS.

Dataset	Video	Length by second	Width×Height	Size	Execution time
University	Video1	10	1920×1027	8.29MB	156s
	Video2	11	1280×720	10.9MB	184s
	Video3	14	1920×1027	11.2MB	254s
School	Video1	16	1920×1080	108MB	392s
	Video2	12	1920×1080	80MB	388s
	Video3	1:16	720×1280	30.5MB	660s
Actors	Video	1:06	720×1280	11.1MB	532s

V. CONCLUSION AND FUTURE WORK

This paper propose an attendance recording system based on face detection and recognition models called Yunet, Sface respectively which excellently outperforms traditional techniques in the mechanism of deep face detection and recognition in low hardware equipment. It is concluded that Yunet and Sface models have a significant impact on solving the problems of face detection and recognition in tough conditions were both are common with in the property of five landmarks considering. However, the proposed powerful and effective deep learning system by three stages for face detection and recognition designed to handle classrooms attendance in educational environments with excellent results. Future work is to applying the system in real-time and executing in different operating system such android for mobile to facilitates life for academics' people.

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